Lark

A daily companion for high-frequency hearing loss

Bring the top of the sound back, and protect what you still have.

Focus area: Healthcare (preventative + everyday wellbeing) · Format: Mobile app, with a wearable / hearing-aid companion · Audience: Global

1. Focus area and the problem

Lark sits in preventative and everyday healthcare. It addresses **high-frequency (sensorineural) hearing loss**: reduced hearing in the roughly 2,000 to 8,000 Hz band. This is not a loss of volume but a loss of clarity. Vowels stay loud, so speech still sounds like speech, but the quiet, high-pitched consonants that carry meaning (s, f, th, sh, k, t) thin out or vanish. The result is the hallmark complaint audiologists hear most: “I can hear, but I can’t understand.”

The condition hides for years because the loud part of speech survives, so people compensate without realising anything is wrong. It is also striking the young: the WHO estimates over a billion people aged 12 to 34 are at risk of permanent hearing loss from unsafe listening through personal devices and loud venues. The fact that reframes the whole product is this: high-frequency loss is, in almost all cases, **permanent**. It cannot be reversed. Flipped around, that is the most motivating fact in the design: every decibel you protect today, you keep. Lark turns a permanent diagnosis into a daily practice of noticing, understanding, and protecting.

2. Target user

Primary: the late-noticer. An adult in their 20s to 40s who grew up with headphones and live music, has recently realised they struggle to follow speech in noisy places, and may have just been told (or suspects) they have a high-frequency or “sloping” loss. They feel some regret and a lot of “what now?” They want clarity in conversation, honest information, and a way to stop things getting worse without feeling old or broken.

Secondary users: anyone with sloping high-frequency loss including age-related; musicians and noisy-trade workers; parents of children with high-frequency loss; people living with tinnitus; and, most important for long-term impact, the not-yet-affected young listener whose hearing is still fully protectable.

3. What Lark does (functionalities)

Lark is organised into five capabilities, each answering a specific part of living with the condition.

- **Your Hearing, see what you can’t feel.** Your high-frequency profile rendered as a “sound horizon”: an audiogram reimagined as a landscape, with consonants and everyday sounds (birdsong, alarms) plotted where they live, so you can see what you are missing. Includes an at-home screening check, framed as awareness rather than diagnosis, that nudges you toward a professional extended high-frequency and speech-in-noise test.
- **Clarity, understand, not just hear.** The everyday hero. Live, on-device captions that emphasise the high-frequency consonants you lose, with a “clarity boost” that visually (and, when paired with hearing devices, audibly) lifts the high band. A minimal-pairs helper disambiguates near-misses such as “sip, ship, or sit?”, and a noisy-room mode focuses on the speaker.
- **Listen-Train, rebuild discrimination.** Short, calm daily exercises that train your brain to tell apart the consonant pairs that blur together (s/f, th/s, sh/s, k/t). Grounded in evidence that listening training

lowers the cognitive effort of understanding speech over time. Progress is shown as listening effort trending down.

- **Protect, keep what you have.** The agency engine. A live sound-dose meter turns the day's noise exposure into a simple budget ("you've used 62% of today's safe sound"), warns when headphones or a venue cross into damage, coaches the 60/60 rule (60% volume, 60 minutes, then a break), and tracks a protected-listening streak.
- **Catch, never miss the high sounds.** On-device recognition of the high-pitched sounds your hearing skips, such as smoke alarms, microwave beeps, the kettle, the doorbell, a baby's cry, or your name, delivered as visual and haptic alerts. And, gently, the one that gives the product its name: a lark is singing nearby.

4. Ethical considerations and mitigations

- **Screener, not a diagnosis.** The biggest risk is someone trusting an app over a clinic. Every self-test is explicitly framed as awareness and repeatedly points the user to a licensed audiologist and the specific tests that catch early loss. Lark never claims to diagnose or treat.
- **No manufactured health anxiety.** For a condition wrapped in regret and stigma, framing matters. Copy is honest but non-catastrophising, and the product always pairs a problem with an action the user can take.
- **The microphone is sacred.** Clarity and Catch require listening to the environment, the most invasive permission there is. Mitigation: all audio processing happens on-device, audio is never recorded, stored, or uploaded, the dosimeter measures level not content, and the mic state is always visibly indicated with one-tap off.
- **Recognition bias and accuracy.** Captions and sound detection can fail across accents, languages, and acoustic settings, and a missed alarm is a safety issue. Mitigation: conservative, clearly labelled confidence; critical alerts such as smoke alarms tuned for high recall; no over-promising; and continual evaluation across voices and languages.
- **Complement, not replace.** Lark supports hearing aids, audiologists, and protective equipment. It does not pretend to substitute for them.
- **Equity.** Core protective features (dose meter, education, alerts) stay free, because the people most exposed to noise often have the least access to care.

5. Privacy and security

Hearing data is a permanent, identifying record of a person's body. It deserves the strongest protection in the product.

- **On-device by default.** Live audio for captions, dosimetry, and sound detection is processed locally and discarded. It never leaves the phone.
- **No audio recordings, ever,** without separate, explicit, revocable consent, and never for advertising or sale.
- **You own your audiogram.** Your hearing profile is encrypted, exportable, and deletable. You choose whether to share it with an audiologist or your hearing devices.
- **Data minimisation and transparency.** The app keeps the least it needs (a daily dose number, not a log of what you heard), with plain-language consent, granular permissions, and a visible microphone indicator at all times.

6. UI mock-up and design tool

The accompanying mock-up presents all five capabilities across an interactive five-tab phone interface (Hearing, Clarity, Train, Protect, Catch), with annotated design notes beside each screen. It was built as a **high-fidelity, interactive prototype hand-coded in HTML, CSS, and SVG**, rather than in a visual design tool such as Figma, Sketch, or Adobe XD. It opens and runs in any web browser, and every tab and control is live.

Design and accessibility principles:

- Show, don’t sound: every alert and status is visual and haptic first, since the product can never assume you will hear it.
- Lower the cognitive load: large legible type, generous spacing, one idea per screen, captions everywhere, and calm pacing, so the interface reduces listening effort rather than adding to it.
- Make the invisible visible: the recurring “horizon” language lets people finally see their own hearing and what they are missing.
- Calm, never alarmist: a naturalist, dawn-toned palette and warm copy, with WCAG-aligned contrast, scalable text, and dark and light modes.